INSTRUMENT CLUSTER (Continued)

The timer function of the EMIC will also automatically turn the rear window defogger system Off. The timer turns the system Off after about ten minutes of operation; however, after the first timed interval has expired, each time the system is turned On again during that same ignition cycle, the timer will automatically turn it Off after about five minutes of operation.

The hard wired input and output of the EMIC rear window defogger control may be diagnosed using conventional diagnostic methods; however, there are no other diagnostic tools available for the EMIC rear window defogger timer and logic circuitry. If the input and output components and circuits of the rear window defogger system test OK, but the system fails to operate as designed, the EMIC must be replaced.

INPUT AND OUTPUT CIRCUITS

HARD WIRED INPUTS

The hard wired inputs to the EMIC include the following:

- **ABS Warning Indicator Driver**
- **Brake Warning Indicator Sense**
- **Driver Door Ajar Switch Sense**
- **Fog Lamp Feed**
- **Front Axle Lock Request (w/Off-Road Package Only)**
- **Front Lock Indicator Switch Sense (w/Off-Road Package Only)**
- **Fused B(+) - Ignition-Off Draw**
- **Fused Ignition Switch Output (Run-Start)**
- **Headlamp Switch Output**
- **High Beam Indicator Driver**
- **Key-In Ignition Switch Sense**
- **Left Turn Signal**
- **Panel Lamps Dimmer Signal**
- **Passenger Door Ajar Switch Sense**
- **Rear Axle Lock Request (w/Off-Road Package Only)**
- **Rear Lock Indicator Switch Sense (w/Off-Road Package Only)**
- **Rear Window Defogger Switch Sense**
- **Right Turn Signal**
- **Seat Belt Switch Sense**
- **4WD Switch Sense (w/o Off-Road Package only)**

Refer to the appropriate wiring information for additional details.

HARD WIRED OUTPUTS

The hard wired outputs of the EMIC include the following:

- **Courtesy Lamp Feed**
- **Axle Lock Switch Enable 1 (w/Off-Road Package Only)**
- **Axle Lock Switch Enable 2 (w/Off-Road Package Only)**
- **Panel Lamp Feed**
- **Rear Window Defogger Relay Control**

Refer to the appropriate wiring information for additional details.

GROUNDS

The EMIC receives a ground path through the following hard wired circuit:

- **Ground - G201**

Refer to the appropriate wiring information for additional details.

COMMUNICATION

The EMIC has the following data bus communication circuit:

- **PCI Data Bus**

Refer to the appropriate wiring information for additional details.

DIAGNOSIS AND TESTING - INSTRUMENT CLUSTER

If all of the instrument cluster gauges and/or indicators are inoperative, refer to PRELIMINARY DIAGNOSIS. If an individual gauge or Programmable Communications Interface (PCI) data bus message-controlled indicator is inoperative, refer to ACTUATOR TEST. If an individual hard wired indicator is inoperative, refer to the diagnosis and testing information for that specific indicator. If the instrument cluster interior lighting control function is inoperative, refer to INTERIOR LIGHTING CONTROL DIAGNOSIS. If the axle locker control is inoperative, refer to AXLE LOCKER CONTROL DIAGNOSIS. If the instrument cluster rear window defogger control function is inoperative, refer to REAR WINDOW DEFOGGER CONTROL DIAGNOSIS.

Refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, details of wire harness routing and retention, connector pin-out information and location views for the various wire harness connectors, splices and grounds.